RODRIGO DE POOL

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https://rodrigodepool.github.io/

ACADEMIC POSITIONS

University of Notre Dame du Lac — **Indiana, USA**
Society of Science Postdoctoral Fellow. — *2025 - 2028*

Instituto de Ciencias Matemáticas — **Madrid, Spain**
Predoctoral Researcher for the Spanish National Research Council. — *2021 - 2025*
Ph.D. in Geometry and Algebra.

- Advisor: Prof. Javier Aramayona
- Research area: Mapping class groups & Teichmüller theory

EDUCATION

Universidad Autónoma de Madrid — **Madrid, Spain**
Master in Mathematics and Applications — *2020 - 2021*

- Master thesis: Modular groups and arc complexes **(10/10)**
- GPA: 9.64/10

Universidad Autónoma de Madrid — **Madrid, Spain**
Double degree in Mathematics and Computer Science — *2015 - 2020*

- Final year project for Mathematics: Classification of surfaces **(10/10)**
- Awarded 29 *Matrículas de Honor* (Highest distinction).
- GPA: 9.30/10

PUBLICATIONS

1. Homomorphisms between pure mapping class groups.
   Advances in Mathematics (2025).
2. Non-planar ends are continuously unforgettable.
   Joint with J. Aramayona, R. Skipper, J. Tao, N. G. Vlamis and X. Wu.
   International Mathematics Research Notices (2025).
3. Braided multitwists.
   Algebraic and Geometric Topology (2025).
4. Finite rigid sets of the non-separating curve complex.
   Forum Mathematicum (2022).
5. Matching arc complexes: Connectedness and hyperbolicity.
   Joint with J. Aramayona and A. Fernández.
   Expositiones Mathematicae (2021).
6. Towards the Automatic and Schedule-Aware Alerting of Internetwork Time Series.
   Joint with D. Perdices, J. García-Dorado, J. Ramos, J. Aracil.
   IEEE Access (2021).

PREPRINTS

1. Rigidity of maps between configuration spaces.
   Joint with P. Huxford, D. Minahan and J. Schillewaert.
   In arXiv:2607.05826 (2026).
2. Holomorphic maps between moduli spaces II.
   Joint with J. Souto.
   In arXiv:2412.01257 (2024).

RESEARCH STAYS

Institut de Recherche Mathématique de Rennes — **Rennes, France**
Collaboration with prof. Juan Souto — *April - June 2024*

Institut Henri Poincaré — **Paris, France**
Scientific Program: — *April - June 2022*
Groups acting on Fractals, Hyperbolicity and Self-Similarity

AWARDS, FELLOWSHIPS, & GRANTS

Premio de Investigación Matemática Vicent Caselles — **2026**
Royal Spanish Mathematical Society and BBVA Foundation

Provost Society of Science Fellowship — **2025 - 2028**
University of Notre Dame du Lac.

FPI predoctoral contract — **2021 - 2025**
Spanish Ministry of Science and Innovation.

JAE Intro SOMdM Grant — **2020 - 2021**
Research grant awarded by Spanish National Research Council.

Excellence scholarships. — **2015 - 2020**
Awarded yearly by the Community of Madrid.

INVITED TALKS, SEMINARS & COLLOQUIA

Universidad Complutense de Madrid — **October 20, 2026**
Polynomials, configurations and braids.

Rice University — **October 5, 2026**
Polynomials, configurations and braids.

University of Notre Dame — **September 1, 2026**
Braid groups and Configuration spaces.

Wasatch Topology Conference — **August 20, 2026**
Polynomials, configurations and braids.

Redbud Topology Conference — **April 11, 2026**
Rigidity of moduli spaces and mapping class groups.

University of Notre Dame **September 23, 2025**
Rigidity results for the mapping class group.

Instituto de Ciencias Matemáticas **May 22, 2025**
Resultados de rigidez del grupo modular.

Universidad Nacional Autónoma de México **October 28, 2024**
Braided elements & homomorphisms in the mapping class group.

Universidad Complutense de Madrid **September 23, 2024**
Homomorphisms between mapping class groups.

Internationales Wissenschaftsforum Heidelberg **July 8, 2024**
Transverse measures to infinite type laminations.

Institut de Recherche Mathématique de Rennes **June 24, 2024**
Holomorphic rigidity of moduli space.

Institut de Recherche Mathématique de Rennes **June 19, 2024**
Mapping class groups, braided elements and homomorphisms.

Instituto de Ciencias Matemáticas **December 5, 2023**
Braided multitwists in the mapping class group.

Universidad de León **February 9, 2023**
Finite rigidity of non-separating curve complexes.

Universidad Autónoma de Madrid **December 14, 2022**
Introduction to geometric group theory.

Instituto de Ciencias Matemáticas **December 2, 2022**
The symplectic representation of the mapping class group.

Institut Henri Poincaré **June 2022**
Finite rigid sets of the non-separating curve complex.

Instituto de Ciencias Matemáticas **May 24, 2022**
The curve complex.

Instituto de Ciencias Matemáticas **December 9, 2021**
On the hyperbolicity of arc complexes and some of their relatives.

TEACHING EXPERIENCE

Instructor for Calculus A **Fall Term 26/27**
University of Notre Dame du Lac.

Instructor for Calculus B **Spring Term 25/26**
University of Notre Dame du Lac.

Teaching Assistant for Mathematical Analysis **1st Term 24/25**
Mathematics, Universidad Autónoma de Madrid.

Teaching Assistant for Linear Algebra **1st Term 23/24**
Data Science and Engineering, Universidad Autónoma de Madrid.

Teaching Assistant for Mathematics **1st Term 22/23**
Enviromental Science, Universidad Autónoma de Madrid.

Teacher in the 'Pequeño instituto matemático' (PIM) **2022 - 2024**
PIM is a weekly activity organized by the ICMAT where high-school students discuss math Olympiad style problems.

CONFERENCE ORGANIZATION

Geometric Group Theory and Low-dimensional Topology **May 2022**
Co-organizar of Graduate School at Instituto de Ciencias Matemáticas.

WORKSHOP & CONFERENCE PARTICIPATION

Wasatch Topology Conference **August 2026**
University of Utah and Bringham Young University. *Utah, USA*

Advanced School in Geometric Group Theory Lluis Santaló **July 2026**
Universidad Internacional Menéndez Pelayo. *Santander, Spain*

Advanced school: Aspherical 2-complexes and their fundamental groups **June 2026**
Instituto de Ciencias Matemáticas. *Madrid, Spain*

Young Geometric Group Theory XIV **June 2026**
Korea Institute for Advanced Study. *Seoul, South Korea*

Curve counting: What's next? **May 2026**
Hamilton Mathematics Institute. *Dublin, Ireland*

Redbud Topology Conference **April 2026**
University of Oklahoma. *Oklahoma, USA*

A conference on Geometry, Topology and Dynamics of Infinite-type surfaces **June 2025**
Casa Matemática Oaxaca. *Oaxaca, Mexico*

Young Geometric Group Theory XIII **April 2025**
University of Copenhagen. *Copenhagen, Denmark*

Group, Topology and Applications VI **April 2025**
University of the Basque Country. *Bilbao, Spain*

Red temática de geometría y física **November 2024**
Instituto de Ciencias Matemáticas. *Madrid, Spain*

Groups in Madrid **October 2024**
Instituto de Ciencias Matemáticas. *Madrid, Spain*

Geometry and dynamics of surfaces beyond finite type **July 2024**
Heidelberg University. *Heidelberg, Germany*

Young Geometric Group Theory XII **April 2024**
University of Bristol. *Bristol, UK*

Groups in Madrid **October 2023**
Instituto de Ciencias Matemáticas. *Madrid, Spain*

Group Actions and Low-dimensional Topology **July 2023**
Organized by Instituto de Ciencias Matemáticas. *Barco de Ávila, Spain*

Groups and Dynamics in Geometry **June 2023**
Organized by ETH Zürich. *Ascona, Switzerland*

Young Geometric Group Theory XI **February 2023**
University of Münster. *Münster, Germany*

Young researchers congress of the Real Sociedad Matemática Española **February 2023**
Universidad de León. *León, Spain*

Groups, Topology and Applications IV **February 2023**
University of the Basque Country. *Bilbao, Spain*

Groups in Madrid **October 2022**
Instituto de Ciencias Matemáticas. *Madrid, Spain*

Groups acting on Fractals **April 2022**
Centre International de Rencontres Mathématiques. *Luminy, France*

Biannual meeting of the Real Sociedad Matemática Española **January 2022**
University of Castilla-La Mancha *Ciudad Real, Spain*

Groups in Madrid **November 2021**
Instituto de Ciencias Matemáticas. *Madrid, Spain*